

1 REFLX: REACTION ENHANCEMENT IN FITNESS
2 USING LIGHT-BASED EXERCISES FOR UPV ATHLETES

3 A Special Problem
4 Presented to
5 the Faculty of the Division of Physical Sciences and Mathematics
6 College of Arts and Sciences
7 University of the Philippines Visayas
8 Miag-ao, Iloilo

9 In Partial Fulfillment
10 of the Requirements for the Degree of
11 Bachelor of Science in Computer Science by

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Approval Sheet

The Division of Physical Sciences and Mathematics, College of Arts and
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certifies that this is the approved version of the following special problem:

**REFLX: REACTION ENHANCEMENT IN FITNESS
USING LIGHT-BASED EXERCISES FOR UPV ATHLETES**

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Declaration

We, Christian Dale P. Celestial and Mark Rensel D. Yeban, hereby certify that this Special Problem has been written by us and is the record of work carried out by us. Any significant borrowings have been properly acknowledged and referred.

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Dedication

“Hello, world.”

Acknowledgment

“Hello, world.”

Abstract

From 150 to 200 words of short, direct and complete sentences, the abstract should be informative enough to serve as a substitute for reading the entire SP document itself. It states the rationale and the objectives of the research. In the final Special Problem document (i.e., the document you'll submit for your final defense), the abstract should also contain a description of your research results, findings, and contribution(s).

Suggested keywords based on ACM Computing Classification system can be found at https://dl.acm.org/ccs/ccs_flat.cfm

Keywords: Keyword 1, keyword 2, keyword 3, keyword 4, etc.

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Chapter 1

Introduction

1.1 Overview of the Current State of Technology

In the realm of sports science, the enhancement of athletes' response time and agility remains a critical focus, as these attributes directly influence performance in dynamic, unpredictable environments such as team sports and combat disciplines (Hassan et al., 2023). Traditional training methods, including cone drills and ladder exercises, have long been employed to improve these skills, yet they often fall short in replicating the rapid, stimulus-driven demands of real-game scenarios. Over the past decade, light-based reaction training systems—devices utilizing visual stimuli like LED lights to prompt immediate motor responses—have emerged as innovative tools to bridge this gap. Systems such as FITLIGHT, BlazePod, and XLiGHT have been critically analyzed for their design features, including sensor connectivity, battery life, and operational reliability, revealing strengths in portability and customization but limitations in diagnostic precision

84 and validity (Ezhov et al., 2021).

85 Empirical studies have demonstrated that these systems can significantly enhance
86 visual-motor coordination, reaction speed, and cognitive functions. For instance,
87 interventions using FITLIGHT in small-sided games have led to marked improve-
88 ments in harmonic abilities (e.g., rhythmization and responsiveness) and basic
89 skills like dribbling among young basketball players (Hassan et al., 2023). Simi-
90 larly, a 10-week FITLIGHT program improved reaction times and dribbling speeds
91 in female basketball athletes, with effect sizes indicating substantial neural adapta-
92 tions (Hassan, 2025). In motorsport contexts, light-based reactive agility training
93 has boosted selective attention, cognitive flexibility, and cardiorespiratory capac-
94 ity in car racing drivers (Horváth et al., 2022). A systematic review of visual
95 training interventions, including light board and stroboscopic methods, further
96 corroborates these benefits, reporting 5-27% reductions in reaction time across
97 various sports, with greater efficacy in elite and younger athletes (Jothi et al.,
98 2025). Reliability assessments of systems like BlazePod have also affirmed their
99 validity for measuring simple and complex reactions in mixed martial arts (MMA)
100 athletes, with high intraclass correlations supporting their use in training proto-
101 cols (Polechoński et al., 2024).

102 Despite these advancements, significant gaps persist in the literature. While light-
103 based systems show promise in controlled settings, their predictive value for field-
104 based reactive agility remains limited, as evidenced by weak correlations between
105 laboratory reaction speeds and on-field performance in soccer players (Broodryk
106 et al., 2025). This suggests a disconnect between isolated visual stimuli and the
107 multifaceted perceptual-cognitive demands of sports, highlighting the need for
108 more integrated, sport-specific designs. Moreover, comparative analyses under-

109 score inconsistencies in system performance, such as variable Bluetooth stability
110 and sensor delays, which could undermine training reproducibility (Ezhov et al.,
111 2021). These issues are largely tied to the wireless protocols used by commer-
112 cial systems—typically Bluetooth or Wi-Fi—both of which introduce transmis-
113 sion overhead and variable latency, limiting their suitability for real-time reaction
114 measurement. This technological gap highlights the need to explore emerging
115 low-latency communication protocols.

116 1.2 Problem Statement

117 DO NOT FORGET to write the statement of the research problem here, i.e.,
118 before the Research Objectives.

119 A problem statement is your research problem written explicitly. The problem
120 statement should do four things:

- 121 1. Specify and describe the problem (with appropriate citations)
- 122 2. Provide evidence of the problem's existence
- 123 3. Explain the consequences of NOT solving the problem
- 124 4. Identify what is not known about the problem that should be known.

1.3 Research Objectives

1.3.1 General Objective

This subsection states the over-all goal that must be achieved to answer the problem. Address the following: Given your research challenge or opportunity, how do you intend to solve it? What is the output of your research?

1.3.2 Specific Objectives

This subsection is an elaboration of the general objective. It states the specific steps that must be undertaken to accomplish the general objective. These objectives must be **S**pecific, **M**easurable, **A**ttainable, **R**ealistic, **T**ime-bounded. A specific objective start with “to <verb>” for example: to design/survey/review/analyze.

Studying a particular programming language or development tool (e.g., to study Windows/Object-Oriented/Graphics/C++ programming) to accomplish the general objective is inherent in all thesis and, therefore, must not be included here.

1. To review related literature, compare and contrast existing algorithms (on what problem?);
2. To develop a new algorithm (for what purpose?)
3. To analyze the algorithm (based on what criteria?)

1.4 Scope and Limitations of the Research

This section discusses the boundaries (with respect to the objectives) of the research and the constraints within which the research was developed.

1.5 Significance of the Research

This section explains why research was done in this area. It rationalizes the objective of the research with that of the stated problem. Avoid including sentences such as “This research is beneficial to the proponent/department/college” as this is already an inherent requirement of all BSCS majors. Focus on the research’s contribution to the Computer Science field.

The following are guide questions that may help your formulate the significance of your research.

- What is the relevance of your work to the computer science community?
 - What are your technical contributions, in terms of algorithms, or approaches, or new domain?
 - What is your value-added compared to existing systems?
- What are your contributions to society in general?
 - Who benefits from your system?
 - Who are your target users and how this system benefit them?

Chapter 2

Review of Related Literature

This chapter discusses the features, capabilities, and limitations of existing research, algorithms, or software that are related/similar to the Special Problem.

The reviewed works and software must be arranged either in chronological order, or by area (from general to specific). Observe a consistent format when presenting each of the reviewed works. This must be selected in consultation with the adviser.

DO NOT FORGET to cite your references.

A literature review must do these things:

- be organized around and related directly to the thesis or research question you are developing
- synthesize results into a summary of what is and is not known
- identify areas of controversy in the literature

- formulate questions that need further research

A literature review is a piece of discursive prose, not a list describing or summarizing one piece of literature after another. It's usually a bad sign to see every paragraph beginning with the name of a researcher. Instead, organize the literature review into sections that present themes or identify trends, including relevant theory. You are not trying to list all the materials published, but to synthesize and evaluate them according to the guiding concept of your thesis or research question. You should also state the limits or gaps of their researches wherein you will try to fill these gaps in accordance to your research problem and objectives.

2.1 Theme 1 Title

This chapter contains a review of research papers that:

- Describes work on a research area that is similar or relevant to yours
- Describes work on a domain that is similar or relevant to yours
- Uses an algorithm that may be useful to your work
- Uses a software / tool that may be useful to your work

It also contains a review of software systems that:

- Belongs to a research area similar to yours
- Addresses a need or domain similar to yours
- Is your predecessor

192 **2.2 Theme 2 Title**

193 **2.3 Chapter Summary**

194 Should include a table of related studies comparing them based on several criteria.

195 Highlight research gaps and the research problem.

Chapter 3

Research Methodology

This chapter lists and discusses the specific steps and activities that were performed to accomplish the project. The discussion covers the activities from pre-proposal to Final SP Writing.

3.1 Research Activities

Research activities include inquiry, survey, research, brainstorming, canvassing, consultation, review, interview, observe, experiment, design, test, document, etc. Be sure that for each method, process, or algorithm used, there is a justification why that method was chosen. The methodology also includes the following information:

- who is responsible for the task
- the resource person to be contacted

- 209 • what were done
- 210 • when and how long the activity was done
- 211 • where it was done
- 212 • why should the activity was done

213 Chapter 4

214 Results and Discussions

215 This chapter presents the results or the system of your SP. Include screenshots,
216 tables, or graphs and provide the discussion of results.

217 **Chapter 5**

218 **Conclusion**

219 This chapter summarizes your SP and provides conclusions regarding your results
220 and analyses. Provide recommendations on what ought to be done with your SP
221 or provide further directions on the topic you covered.

Chapter 6

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²⁵⁴ **Appendix A**

²⁵⁵ **Code Snippets**

256 Appendix B

257 Resource Persons

258 **Dr. Firstname1 Lastname1**

259 Role1

260 Affiliation1

261 emailaddr@domain.com

262 **Mr. Firstname2 Lastname2**

263 Role2

264 Affiliation2

265 emailaddr2@domain.com

266 **Ms. Firstname3 Lastname3**

267 Role3

268 Affiliation3

269 emailaddr3@domain.net

270